Simultaneous Acquisition and Sensing of ECG and IPG Signals for measurement of Pulse Transit Time

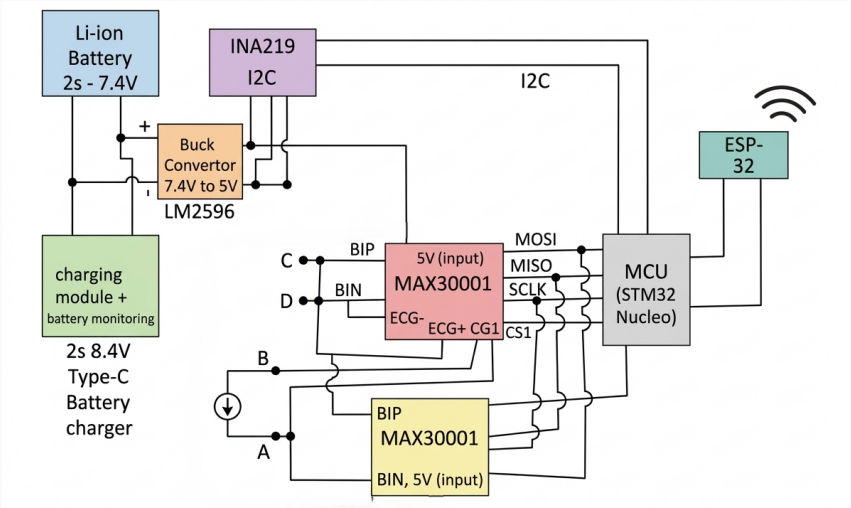
Group MON-04 — Milestone 2

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IIT Bombay

Block diagram



Battery Management System (BMS) Changes

- **Battery Configuration Switched: 1S $\rightarrow$ 2S Lithium Pack**
  - **Reason 1 (Component):** MAX30001 Breakout Board requires a stable 5V supply.
  - **Reason 2 (Power):** Higher input voltage ensures proper LDO dropout voltage for stable regulation.
  - **Output:** Provides 7.4V (Nominal) / 8.4V (Max).
- **Integrated Charging and Protection Module**
  - **Module:** Type-C USB 2S BMS Charging Boost Module (8.4V / 12.6V).
  - **Key Features:**
    1. Supports **2S balanced charging**.
    2. Includes **Integrated BMS Circuit** for:
    3. Over-charge and over-discharge protection.
    4. Short-circuit protection.
    5. Cell balancing.

Monitoring, Regulation, and PCB Components

- **Battery Monitoring**
  - **Component:** INA219 Current/Voltage Sensor Module.
  - **Function:** Measures battery voltage and current.
  - **Communication:** I^2C to the microcontroller.
- **Voltage Regulation**
  - **Component:** LM2596S Buck Converter Module.
  - **Function:** Converts $6V - 8.4V$ battery voltage to a stable **5V output** for system power.
- **PCB Schematic Overview**
  - **Main Headers:** Pin headers for **two MAX chips**, **STM microcontroller**, and **ESP32-DEVKIT-V1** wireless module.
  - **Auxiliary Headers:** Additional **GPIO pin headers** for future testing, debugging, and peripherals.
  - **Debug Features:**
    - **Power LED** to confirm successful power delivery.
    - **Two Debug LEDs** connected to MAX chip select (CS) lines for visual communication identification.

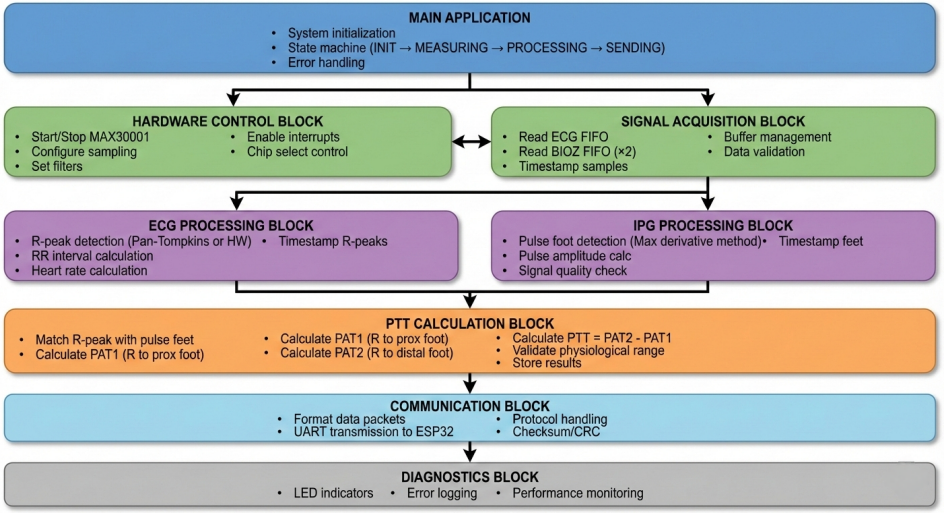
Software Design

Some Design Decisions and Rationale:

- **RTOS:** ensures deterministic sampling (ECG 256 Hz, IPG 128 Hz), ms-accurate timestamps for PTT, and priority-based execution. Thus, acquisition and processing always preempt communication and UI tasks.
- **RTOS Choice:** FreeRTOS selected over Zephyr for lower RAM/Flash usage, simpler STM32 integration, and sufficient real-time features without unnecessary complexity.
- **Driver Design:** Small SPI data transfers handled using interrupt-driven drivers instead of DMA to reduce complexity.
- **Hardware Debugging:** Logic analyzer used to verify SPI timing, chip-select sequencing, and interrupt behavior.

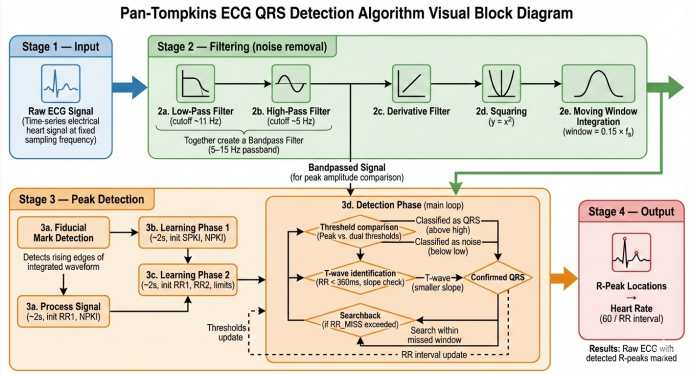
Firmware Functional Block Diagram

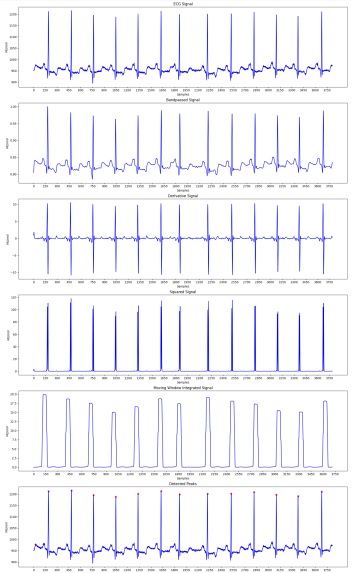
The following diagram presents the block diagram of the code, along with their features.



ECG R-Peak Detection (Pan–Tompkins)

The Pan–Tompkins algorithm uses a **linear DSP pipeline** for R-peak detection which combines the **bandpass filtered signal** and the **integrated signal**.





APW Foot Detection: Cubic Polynomial Method.Here is :Reference link

Search Interval

The foot point is assumed to lie between the ECG R–peak (start of cardiac cycle) and the point where the APW has its maximum rising slope.

Algorithm

1. Divide the interval into 10 segments.
2. Approximate each segment using a straight line with least–squares fitting.
3. Compute average time points from neighbouring segments representing the APW foot region.
4. Fit these points with a cubic polynomial.
5. The minimum of this curve gives the APW foot point.

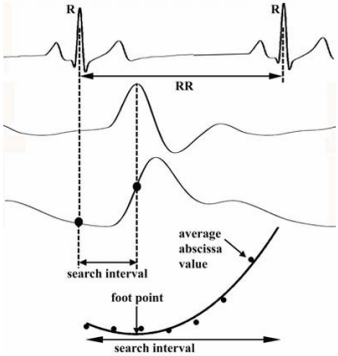


Figure 5. ECG (a), the first derivate of the APW (b), APW (c), zoomed search interval (d)

At SNR = 18.19, error dispersion is ±6 ms (cubic polynomial) being the most accurate method.

PTT Calculation Approach, Reference Links: [1] [2]

Need for ECG in PTT Calculation

PTT can be obtained by:

- Δt between **ECG R-peak** and **BioZ foot**(PAT2-PAT1)
- Δt between **two BioZ waveform foot points**(Direct calculations of PTT)

Foot Detection Algorithms

Second Derivative Method: ECG is not required; the foot is detected at the maximum of the second derivative of the BioZ waveform, with error dispersion ± 14 ms for SNR $= 18.19$.

Tangent Intersection Method: Requires ECG waveform; the foot is estimated from the intersection of two tangents , resulting in error dispersion of about ± 11 ms.

Cubic Polynomial Method: Requires ECG waveform; a cubic polynomial is fitted to a larger segment of the BioZ waveform around the foot, giving the best accuracy with error dispersion of about ± 6 ms.

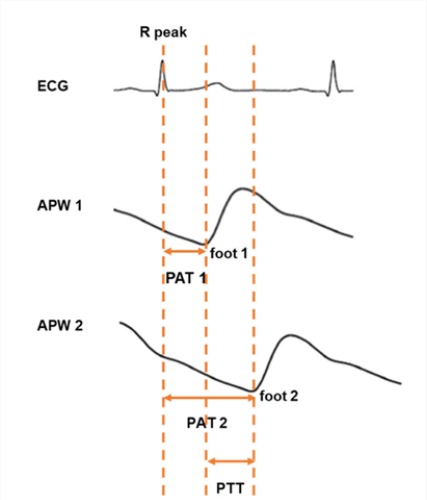
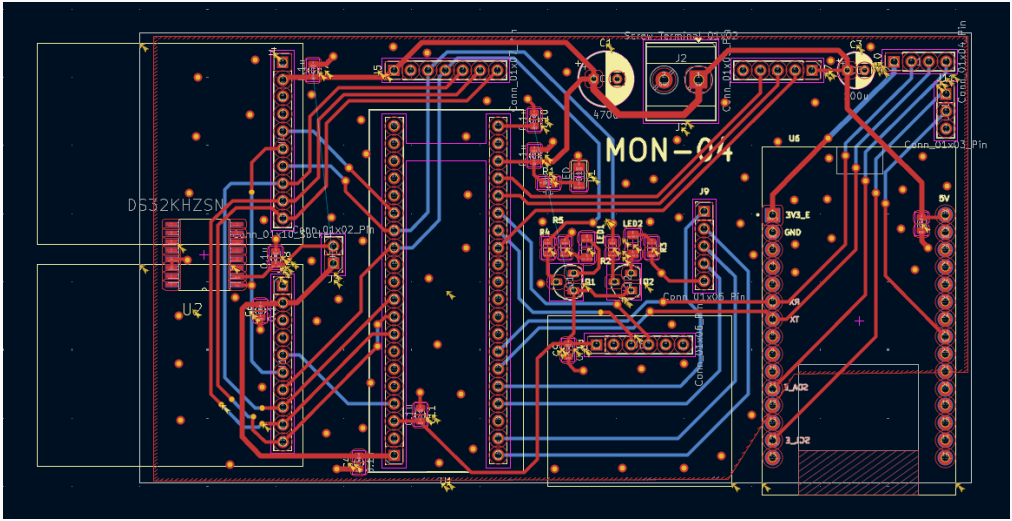


Fig. 1. PTT measured from two PATs between an ECG and two APWs.

PCB Layout

1. We made the PCB layout under the constraints of the IITB PCB LAB fabrication process.
2. It was our preferred choice because it adequately meets our requirements, and we can remain more informed of any issues that might arise during manufacturing.
3. We required a 2-layer PCB, with the top layer housing the SMD components and a ground plane, while the bottom layer only contains a ground plane.
4. We also incorporated feedback to breakout pins corresponding to voltages such as 5V, 3.3V, and GND, serial communication lines, as well as GPIOs to help us with debugging.
5. Decoupling capacitors were placed close to the power input pins to suppress high-frequency noise.

PCB Layout


MON-04
D532KHZSN
U2
U5
U6
J2
J9
Screw_Terminal_01x02
Conn_01x02_Pin
Conn_01x03_Pin
Conn_01x04_Pin
Conn_01x05_Pin
Conn_01x06_Pin
LED2
3V3_E
GND
5V
X1
X2
3_A2
3_D2
4700
100

Risk Management & Mitigation Strategy

Risk Factor	Technical Impact	Mitigation Strategy
Clock Drift / Sync Error	Jitter between dual MAX30001 boards leads to PTT inaccuracy.	Use master-slave `INT` trigger for sample alignment.
Shared-Electrode Interference	Excitation current on the same leads as ECG/IPG1 distorts the primary IPG waveform.	Interim software correction using a Moving Average filter; ongoing investigation into hardware decoupling.
AFE Saturation	High skin-electrode contact impedance clipping the ADC.	Real-time Lead-Off Detection (LOD) and hardware Gain/Bias recalibration.
Motion Artifacts	Z changes exceeding the internal DSP recovery window ($>$ 100ms).	Trying to keep the user as still as possible for good signal acquisition.

Common Right Leg electrode is maintained between both boards to prevent DC offset drift.

Chip Interfacing: Low-Level Register Configuration

Configuration Strategy

- System parameters are dynamically tuned via SPI-accessible registers to ensure dual-board synchronization. The simultaneous `REG_SYNCH` command resets internal timers on both chips to ensure aligned sampling across dual data streams.

Active vs. Passive Configuration

1. **Board 1 (Active):** 64 μA current injection ($cgmag[13:11] = 111$).
2. **High-Pass Filter:** 0.5 Hz ($AHPF[2:0] = 010$) on both boards for baseline stability.
3. **Low-Pass Filter:** DLPF bypassed (00) for maximum raw bandwidth.
4. **Board 2 (Passive):** 0 μA injection (000) to prevent crosstalk; 80 V/V gain.

Synchronization: Simultaneous `REG_SYNCH` resets at 1 MHz SPI ensure zero-offset timing between dual IPG streams.

Signal Acquisition: Dual-Stage GUI Averaging

Noise Mitigation Strategy

- Raw acquisition revealed significant baseline disturbances; a dual-stage averaging filter was developed in the Python GUI.

Software Algorithm

1. **Stage 1:** Moving average window suppresses high-frequency artifacts from the 115200 baud serial stream.
2. **Stage 2:** Secondary windowing applied to primary output for baseline stabilization.
3. **Tuning:** Real-time "Smoothness" sliders (range 2–80) allow interactive SNR optimization.
4. **Output:** Synchronous plotting of raw ECG and filtered dual-channel IPG.

Status: Visual clarity is achieved, but absolute fidelity relative to medical-grade benchmarks remains unquantified until clinical tests.

Before Filtering

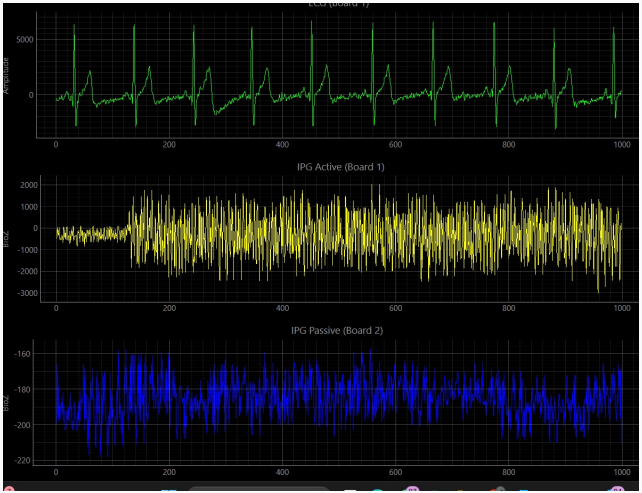


Figure 2: ECG AND BIOZ WAVERFORMS

After Filetring



Figure 3: ECG AND BIOZ WAVERFORMS

Project Timeline Milestone 3 Goals

- **Mar 5–11:** Breadboard validation of the dual-chip acquisition logic.
- **Mar 11–18:** Circuit verification and PCB schematic finalization, along with algorithm implementation on IPG and ECG databases (current phase).
- **Mar 18–25:** Algorithm integration on STM32 and testing on human subjects; tuning parameters to obtain accurate PTT and R–R peak detection, along with PCB soldering.
- **Mar 25–Apr 4:** Development of the GUI interface and final product 3D printing.